Web3 Based Decentralized Crowdfunding Platform Using Blockchain Technology

Ayush Shukla Department of Computer Science and Engineering SRM Institute of Science and Technology, Delhi-NCR Campus, Modinagar, Ghaziabad, U.P., India email: as8786@srmist.edu.in

Pankaj Joshi Department of Computer Science and Engineering SRM Institute of Science and Technology, Delhi-NCR Campus, Modinagar, Ghaziabad, U.P., India email: pj2830@srmist.edu.in

Subhakar Vadlamani Department of Computer Science and Engineering SRM Institute of Science and Technology, Delhi-NCR Campus, Modinagar, Ghaziabad, U.P., India email:sv8534@srmist.edu.in

Aditya Rastogi Department of Computer Science and Engineering SRM Institute of Science and Technology, Delhi-NCR Campus Modinagar, Ghaziabad, U.P., India email: ar7368@srmist.edu.in

Dr. Abilasha Singh Department of Computer Science and Engineering SRM Institute of Science and Technology, Delhi-NCR Campus, Modinagar, Ghaziabad, U.P., India email: abhilass1@srmist.edu.in

Abstract. Most of the existing crowdfunding platforms are operated by centralized companies which dictate all the functionalities of a campaign and investment. This is prone to create the lack of visibility whereby; less transparency is enhanced yet the donor must be confident that the middleman is spending money in a sincere manner [1-3]. To deal with this, researchers are tending to transition to the Blockchain technology which is a permanent and public registry where every and every transaction is publicly recorded and cannot be modified [4], [5]. Ethereum Smart Contracts fully automate the basics of fundraising such as personal donations collection and final funds disbursement. This does the effect of removing the need to have a third-party human manager or central authority [6]-[9]. In this study, the design-based approach was used to develop and test a Web3 prototype on a running blockchain application. We discover that the resultant data indicates that it is much more efficient to put these underlying processes on chain and that it is much easier to audit the result of the system. Ultimately, the findings prove that this decentralized network is far less reliant on intermediaries compared to the conventional platforms that are operational [10], [11].

Keywords: ***Web3, Blockchain, Decentralized Crowdfunding, Smart Contracts, Ethereum, Transparency, Trustless Systems***

1. Introduction

The paper shall define the development of the crowdfunding since and how a decentralized solution to crowdfunding can be achieved using blockchain technology. Crowdfunding is an approach where individuals or organizations gather capital by the joint efforts of many individuals, normally via an online platform, and then the funds collected in the campaign are stored in the account of the intermediary. Such a structure tends to reduce the level of openness, prolongs the process of the transaction, fosters the evolution of an interdependency on the platform, and the increasing level of interdependency on institutional trust [1],[2]. The researchers have also discovered that the contributors within a centralised crowdsourcing system are likely to be exposed to different kinds of risks because they cannot view the history of the transactions and the ultimate decision making activities [3], [4]. Being a transparent, unchangeable history of transactions, blockchain can be used as a replacement of conventional crowdfunding that is decentralized. The shared record will also facilitate the increase of trust and verifiability among all subjects involved [5, 6].

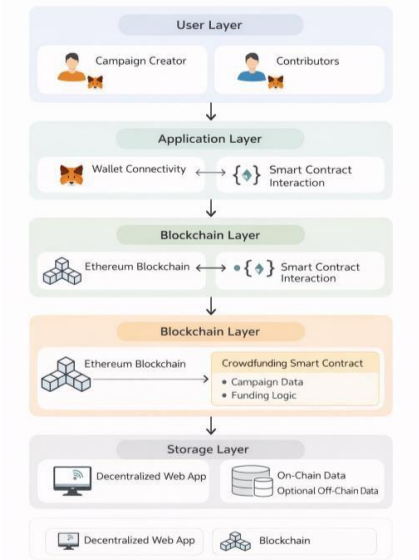


Fig 1. Proposed Web3 Crowdfunding Platform Architecture

Fig 1. shows the general layout of the planned Web 3 platform that will be used to make decentralized crowdfunding. The users will operate the Web3 crowdfunding platform through the use of decentralized web Ethereum Smart Contracts form the core logic of the platform that will process all contributions and audit transactions. This gives information permanence, which is verifiable proof of all the contributions on the Ethereum blockchain [7, 8].

Current technologies within the Web3 have made it possible to develop decentralized applications, which assist in carrying out peer to peer value transfer and transparent financial transactions [9], [10], [11]. In addition, Smart Contracts do not only offer unchangeable records but also allow a smooth process of verification and audit, with

data integrity being ensured [12, 13]. Even though there is a growing amount of literature about blockchain-based crowdfunding, there are still critical research gaps, and technical issues. In spite of the fact that the literature about blockchain based crowdfunding systems keeps growing, nonetheless, some gaps and challenges in the research are observed. Many of the non centralized systems tend to be very technical and thus can possess adverse effects on usability in addition to limiting non technical users [14]. Furthermore, literature says that there are no conventional design frameworks, and there is no extensive based comparisons between the experimental design and the customary centralized crowdfunding sites. Decentralization of crowdfunding is not a common practice yet because it has not undergone real world testing in terms of transparency, speed, and reliability [15]. The loopholes should be sealed to continue with viable, safe and comfortable Web3 based crowdfunding sites.

2. Literature Survey

Current research shows that blockchain technology is becoming more common in financial and transactional systems. Blockchain technology is gaining popularity because it can improve trust, transparency and security. Foundational studies introduced ideas of decentralized and unchangeable records. These enable secure peer to peer transactions without relying on centralized intermediaries [1]. Later research examined consensus mechanisms and blockchain architecture, confirming their effectiveness in ensuring audibility, data integrity and resistance to unauthorized data tampering in distributed networks [2] [3].

Blockchain systems currently rely on smart contracts. They let rules and agreements happen on their own. Early studies suggested that smart contracts diminish the need for trust among users by converting agreements into self-sufficient computer programs [4]. Later, when smart contracts were used in real life on blockchain platforms like Ethereum, it showed that they can automate financial transactions and reduce the need for middlemen. This makes decentralized systems [5], [6] quicker, more dependable, and less prone to mistakes. As a result, smart contracts are now considered a key foundation of decentralized financial and transaction-based applications.
Many studies show that fundraising platforms and blockchain based crowdfunding at most times offer better traceability, transparency and user trust compared to traditional centralized crowdfunding systems. It is suggested that contract-based models automatically handle contributions and manage campaigns which helps in reducing chances of fund misuse by enforcing predefined rules [7], [8]. Blockchain based donation and charity platforms also show how transparent and verifiable tracking of funds can improve accountability [9]. Research on web3 technologies and decentralized applications highlights the role of digital wallets and peer to peer transactions in giving users more control supporting decentralization within online platforms [10] [11].However, existing studies also point out that many proposed systems are still incomplete , mainly theoretical or limited in functionality with very little experimental testing or performance evaluation to prove their effectiveness [12], [13].

Even though there is an increase in research on blockchain based crowdfunding systems, several challenges and research gaps still exist. Many decentralized platforms are technically complex which makes it hard for non technical users to understand and use. [14]. However, this complexity limits their widespread adoption. Studies also show a lack of proper experimental comparisons between traditional centralized crowdfunding platforms and standard design framework.There isn't much real-world testing to see how well the system works, how open knowledge of is, and how trustworthy it is. These problems make people less sure about decentralized crowdfunding solutions and slow down their real-world use [15].Filling up these gaps is important for making web3-based crowdfunding platforms that are easy to use, safe, and broadly recognized.

3. Proposed Model

The proposed system represents a decentralized Web3 based crowd funding platform whose goal is to enhance the security and automation without the need for centralized intermediaries. The framework is built using Blockchain technology which tends to enable smart contracts which allow the project creators and contributors to interact directly without the need of external support. With the integration of crowd funding logic with smart contracts, the financial operations such as campaign creation and fund transfer are automatically executed using the Blockchain technology which promotes the tamper resistance without manual intervention [1], [2].

The design of the system consists of four layers which are user layer, application layer, blockchain layer and storage layer. This layered architecture helps in improving the overall modularity, scalability and separation of concerns which are important component characteristics of a decentralized system [3], [4]. The user layer consists of campaign creators and contributors who interact with the platform through a decentralized web application. Instead of utilizing the conventional methods the user authentication is done through cryptographic wallet integration which allows the participants to securely engage using their blockchain addresses. This technique is useful for protecting user privacy while allowing peer to peer participation and transparency in transaction monitoring [5]. Through the user interface campaign creators define the campaign's specifications and set their funding goals. At the same time the contributors can track the funding progress and transaction history in real time using blockchain records.

The application layer serves the function of a bridge interface between the user facing application and the blockchain network. It establishes a connection between the users crypto wallet, validates the user interaction and communicates with the smart contracts that are already deployed on the blockchain. Essential functions involving campaign initialization, handling contributions and transferring funds are controlled by smart contracts. These ensure automated execution of the rules written in them which prevents unauthorized changes or manipulation.[6],

[7]. The blockchain layer is responsible for maintaining the data of all transactions in a secure and permanent manner. It stores an immutable distributed ledger where every transaction and smart contract activity is kept record off. When a contribution is made

to a campaign the transaction is directly proceeded on the blockchain. All the funds' transfers can be seen and verified publicly on the blockchain ledger. Once a record is confirmed then it cannot be changed and thus it improves the trust and transparency amongst the users [8], [9].
To improve the efficiency while preserving the decentralized system, the storage layer allows for an optional off-chain storage non- important data like campaigning descriptions, images and rest details. Even though at present essential campaign information is directly stored on the blockchain for security and transparency, the system is designed in such a manner that it can support off-chain storage solutions in the future [10]. The hybrid design used in the model helps to reduce the amount of data stored on the blockchain. It also provides the benefits of decentralization with good performance while maintaining integrity within the system.

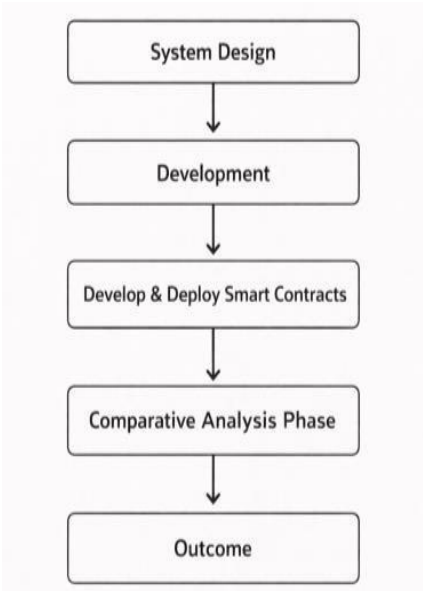


Fig. 2. Workflow of the Proposed Web3 Crowdfunding System

Fig. 2. The figure depicts the step-by-step approach of the design and execution of the Web3 based crowdfunding platform. The process initiates with the system design followed by the development of application and the smart contracts are written and implemented on the blockchain. After deployment, the smart contracts are tested and analyzed following which a comparison is done to evaluate the performance and overall efficiency. In the final stage the results are retrieved to check the effectiveness of the system. This is used to confirm the success and the real world practicality of the proposed approach

During the system design phase the complete structure of the system is designed in accordance with the principles based on Web3 which primarily focuses upon the de-

centralization, transparency and automatic rule enforcement. Important requirements include campaigning creation, allowing contributors to participate with all the main operational logic to run directly on the blockchain so that the system works without having to trust a central authority [11]. In the implementation phase the design is put into implementation. The modular smart contracts are developed and deployed on a blockchain test network. These smart contracts are used for managing campaign registration, processing contributions and funds transfer. This approach provides improved security and thus provides ease of maintenance also facilitating reuse in the future.
During the experimental phase the prototype is tested in a controlled environment to observe the behavior of the system. Essential parameters involving transaction execution time, reliability of smart contracts and overall system response are cautiously examined. The transparency is also evaluated and examined by checking the accessibility and the unchanging nature of the record stored in the blockchain ledger. In the comparative analysis phase, the decentralized crowd funding platform is compared with traditional systems. This is based on the factors such as transparency, dependency on trust and fund security. This analysis is used to showcase the advantages of a blockchain based crowdfunding system in reducing the centralized control and improving the transparency and trust amongst the contributors.

4. Implementation

4.1 System Overview

The system being proposed is a decentralized crowdfunding platform using blockchain technology. This system operates in a decentralized manner unlike the traditional crowdfunding systems that depend upon the centralized servers and databases. It uses Ethereum smart contracts to handle essential tasks including crating campaigns, managing donations and funds transfer. This is crucial for preserving the transparency, security and trustless environment as all the processes happen automatically and thus the user does not depend on any central authority.

The platform has the following components:

- Smart contracts run on the Ethereum Sepolia test network.
- A frontend web application allowing users to interact with the system.
- A metamask wallet login is used for secure authentication.

4.2 Technology Used

4.2.1 Blockchain and Smart Contracts

- Platform used: Ethereum (Sepolia test network)
- Language used: Solidity
- Framework used: Hardhat

- It used proof of stake mechanism

The functions of smart contracts are:

- It is used to handle campaign creation
- Used for the purpose of donations.
- Used for funds transfers and on-chain data storage.

4.2.2 Frontend Specifications

- Framework: React (Vite)
- Styling: Tailwind CSS
- Web3 Library: ethers.js
- Wallet Integration: MetaMask

The frontend connects to the deployed smart contracts using the ABI smart contracts address thus allowing the users to perform and access the blockchain functions directly based on a web browser interface.

4.3 Network Specifications

Test Network: Sepolia Ethereum test network
Reason for usage:-

- Permits free testing using test Ether
- Prevents real time financial loss
- Suitable for academic and experimental use

All the smart contracts deployments and transactions run safely on the Sepolia network. This is a crucial step for ensuring safety during the development and testing.

4.4 Database Usage

No conventional databases such as MySQL, MongoDB or Firebase are used in this system. The blockchain acts as a decentralized data storage system.

4.4.1 On-Chain Data Storage Instead of a traditional database, the system uses blockchain as a decentralized storage system.

The following is the data stored on-chain:

- Campaign details (title, description, target, deadline)
- Campaign owner's wallet address
- Donation amounts
- Wallet address of the donors
- Total funds collected

The data is stored using Solidity structs, mappings and arrays on the blockchain permanently.

4.4.2 Off-chain Storage

- User authentication is done through metamask wallets.
- Transaction records are fetched from the blockchain using RPC calls or Etherscan.
- Images can be stored externally using URLs (IPFS or other hosting)
- The proposed system utilizes the blockchain ledger itself as a decentralized and immutable data store by avoiding the need for a centralized database.

4.5 Wallet - Based authentication

Unlike the conventional login systems the platform uses wallet based authentication.

- Users log in using MetaMask wallet connection
- Each user is identified by Ethereum walled addresses
- All actions are cryptographically signed with the user's private key.
- No username passwords or sensitive data is stored.

This method highly aligns with the Web3 design principles And enhances security of the system.

4.6. Transaction Flow

- Users connect their wallets using MetaMask.
- Frontend sends requests using ethers.js
- Smart contracts execute the requested functions
- The validation is done by the Ethereum networks
- Transaction data is permanently stored on the blockchain

4.7. Security Considerations:

- Only the campaign owners can withdraw funds .
- All transactions are verified by the Ethereum networks.
- Wallet signing maintains secure and authorized actions.

No central server which reduces the risk of hacking.

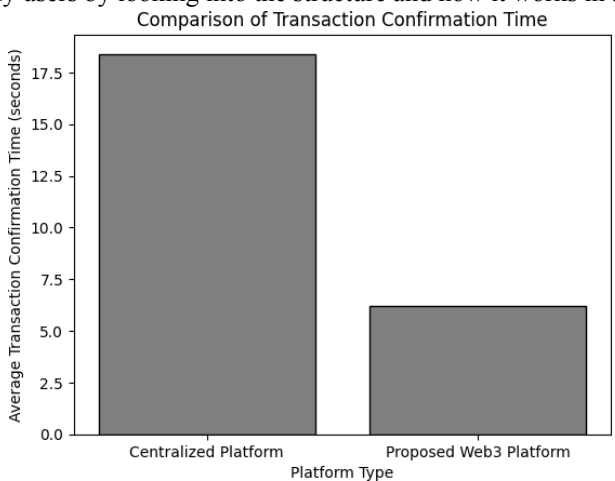
5. Result And Discussion

5.1 Experimental Setup

The proposed Web3-based decentralized crowdfunding platform was tested in a controlled test environment with a blockchain test network known as Sepolia test network. To prevent financial, ethical, and legal issues, real money transactions were not used. Instead, the network gave test tokens and they acted as real crowdfunding actions, allowing safe and proper evaluation of the system without any financial risk.
A working prototype was created, which included smart contracts to manage campaign launch, to control contribution, and check for the milestones. Different unreal crowdfunding campaigns were created to test and run the behavior of the system in different situations, like how it behaves in fully funded campaigns, or partially funded campaigns, or how it behaves in cases where the goal is not achieved.

Metric	**Centralized Platform**	**Proposed Web3 Platform**
Average Transaction Confirmation Time(sec)	18.4	6.2
Transaction Transparency	Limited	Full (On- Chain)
Data Immutability	No	Yes
Transaction Failure Rate (%)	3.6	0.8
Manual Intervention Required	Yes	No
User Authentication Method	Username/ Password	Wallet- Based

Table I. Comparative Performance Metrics Between Centralized and Web3-Based Crowdfunding Platforms
Here, we are checking things like how open the system is, how much it is, and how it is able to verify users by looking into the structure and how it works in a real



situation.

Fig 3. Comparison of Transaction Confirmation Time

Fig 3. This bar graph shows that the proposed web3 decentralized blockchain does the transaction faster than the centralized system. This is because smart contracts work automatically and there is no middleman to slow down things. This shorter time proves that decentralized transactions on the Ethereum test network can be more efficient.

5.2 Performance Evaluation

The study tested three things, the first one is how fast the transaction happens and second is how smart contracts automate tasks and last is how quickly the system responds. The results show that the system with blockchain-based crowdfunding works better and faster than the normal centralized platforms. Creation of the campaigns and the payments runs smoothly because of the smart contract as it handles everything automatically without any manual work. Even though the real money was not used, the system performed like a real blockchain as the test network that we had used followed the same rules as the real ones. Overall, the results show the proposed decentralized platform reduces delays, makes the payments faster and provides a clear and smooth user experience.

5.3 Transparency and Trust Analysis

Transparency and trust were one the important metrics to be judged. It is done by seeing how well the system tracks each and every transaction, and how perfectly the system is working, and how smoothly smart contracts are running. This is all noted without looking at the money profits. While testing every action that we perform, all are permanently recorded on the blockchain. Due to which independent checking was possible for every function like creating the campaigns, contributions and funds which were transferred were matched through public logs. This makes sure that the whole system is trustworthy as users can check all the actions without any central authority control.

Smart contracts are used to make sure that the crowdfunding rules are applied fully, and the system should work automatically, and the power admin should be removed fully. Even though real money was not used the system is build such that it can handle real money as the test network follows the same rules. None of the actions can be changed and it can be monitored in real time. Also each transaction is saved safely and it cannot be deleted or changed at any point of time. Users can check their past actions anytime and the ledger will provide the details to everyone as needed.

5.4 Comparative Analysis

A proper comparison was done between the proposed decentralized crowdfunding platform and traditional centralized crowdfunding systems carefully. In a centralized system mostly all the words like transactions or platform tasks are managed by the middle parties, due to this delays may happen and dependencies on these parties increases by which trust and transparency also decreases on the system. In contrast the

proposed system uses blockchain based smart contracts to run main crowdfunding tasks in a decentralized way such that the central control is reduced.
The result of the test shows that the decentralized approach reduces manual work and lets the system work smoothly through the rules which are defined in the smart contracts. While all the test cases which the real funding platform can have are not tested, the ones which are tested show clear benefits of the system's design and how it works. The system is more open, works automatically, and users don't have to rely too much on a company or middleman to trust that everything is handled fairly.

6. Conclusion

This research paper talks about how to make and build a decentralized crowdfunding platform built on web-3 to fix problems with trust, transparency, and inefficiency that are common in traditional centralized systems. The suggested solution removes centralized control by using smart contracts and blockchain technology, therefore automating tasks such as donation collection, fund transfer, and campaign design. This method makes people more responsible and less dependent on middlemen by making sure that all transactions are clear and stored on the blockchain.

The system is practically possible because a functioning prototype was successfully deployed on a blockchain test network and a layered decentralizing architecture was used.Experimental results show that permanent and interference-proof transaction records, higher transaction efficiency, and less need for manual intervention have all been better recorded than in traditional systems.The study shows that decentralized crowdfunding platforms based on web3 provide a clear, trustworthy, and safe alternative to traditional crowdfunding systems. They also provide a strong foundation for future work and development in decentralized financial applications.

7. Future Scope

The approach that has been suggested already makes things more open, automated, and easy to manage money, but it can still be improved. New features like decentralized governance can assist contributors make decisions, such voting on blockchain to approve project phases or settle disagreements. Connecting separate blockchains could potentially let more people join from other networks. In the future, there may be more user-friendly designs and better standards that decentralized technologies should follow to preserve both user privacy and legal needs. Using the system and testing it for a longer time will also help us learn how people are using it, how it is growing, and how much they trust it. This will help us get it ready for real-world use.

References

1. S. Nakamoto, "Bitcoin: A peer-to-peer electronic cash system," 2008.
2. V. Buterin, "A next-generation smart contract and decentralized application platform," *Ethereum White Paper*, 2014.

3. N. Szabo, “Formalizing and securing relationships on public networks,” *First Monday*, vol. 2, no. 9, 1997.
4. Y. Yuan and F.-Y. Wang, “Blockchain and cryptocurrencies: Model, techniques, and applications,” *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 48, no. 9, pp. 1421–1428, 2018.
5. A. A. Monrat, O. Schelén, and K. Andersson, “A survey of blockchain from the perspectives of applications, challenges, and opportunities,” *IEEE Access*, vol. 7, pp. 117134–117151, 2019.
6. F. Casino, T. K. Dasaklis, and C. Patsakis, “A systematic literature review of blockchain-based applications,” *Telematics and Informatics*, vol. 36, pp. 55–81, 2019.
7. S. Wang, X. Wang, Y. Zhang, et al., “A secure cloud storage framework with access control based on blockchain,” *IEEE Access*, vol. 7, pp. 112713–112725, 2019.
8. W. Teng, H. Xu, Z. Huang, et al., “A smart contract– based service platform for trustworthy crowdfunding,” in *Proc. IEEE ICSS*, 2022.
9. X. Wan, Z. Teng, Q. Li, et al., “Blockchain technology empowers crowdfunding decision-making,” *Expert Systems with Applications*, vol. 221, Art. no. 119685, 2023.
10. M. S. Farooq, M. Khan, and A. Abid, “A framework to make charity collection transparent and auditable using blockchain,” *Computers & Electrical Engineering*, vol. 83, Art. no. 106588, 2020.
11. G. Wood, “Ethereum: A secure decentralized generalized transaction ledger,” *Ethereum Yellow Paper*, 2014.
12. Z. Zheng, S. Xie, H. Dai, et al., “An overview of blockchain technology: Architecture, consensus, and future trends,” in *Proc. IEEE BigData Congress*, 2017.
13. S. S. Hasan, N. H. Sultan, and F. A. Barbhuiya, “Cloud data provenance using IPFS and blockchain technology,” in *Proc. ACM Workshop on Security in Cloud Computing*, 2019.
14. L. Deng, Q. Ye, D. Xu, W. Sun, and G. Jiang, “Determinants of crowdfunding success: A literature review,” *Financial Innovation*, vol. 8, no. 41, 2022.
15. R. Adamo, D. Federico, M. Intonti, et al., “Crowdfunding: Models, risks, and regulation,” in *Banking and Finance*. IntechOpen, 2020.